



UTM
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DATABASE DESIGN DESCRIPTIONS: CONCEPTUAL DATABASE DESIGN

FOR Hasta Travel & Tours Car Rental System

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Table of Contents

Table of Contents	2
1.0 Introduction	3
2.0 DFD (To-Be)	4
3.0 Data And Transaction Requirement	5
3.1 Proposed Business Rule	5
3.2 Proposed Data And Transaction	8
4.0 Database Conceptual Design	12
4.1 Conceptual ERD	12
4.2 Enhanced Erd (EERD)	13
5.0 Data Dictionary	14
6.0 Summary	21

1.0 INTRODUCTION

This report focuses on the Database Conceptual Design for the proposed Hasta Travel & Tours Car Rental Management System. The purpose of this report is to create a clear structured database design based on the requirements gathered. The database supports the system's key features such as customer registration and verification, vehicle management, booking, payment tracking, penalty handling, inspections, and reporting.

Currently, the system still depends heavily on manual processes and communication through WhatsApp to manage rental bookings. Since the information is scattered across chats and different records, it becomes difficult to centralise and track booking history and manage operations. This often leads to delays, human error and redundancy in data. Thus, having a centralized database is important to keep data organized, consistent, and reliable for daily transactions and operation.

A conceptual Entity-Relationship Diagram (ERD) is created to identify the main entities involved in the system, attributes, relationships, and multiplicities. Enhanced ERD (EERD) are applied where suitable to make the design more complete and easier to understand. Overall, this conceptual design will serve as a strong foundation for logical database development and system implementation.

2.0 DFD (TO-BE)

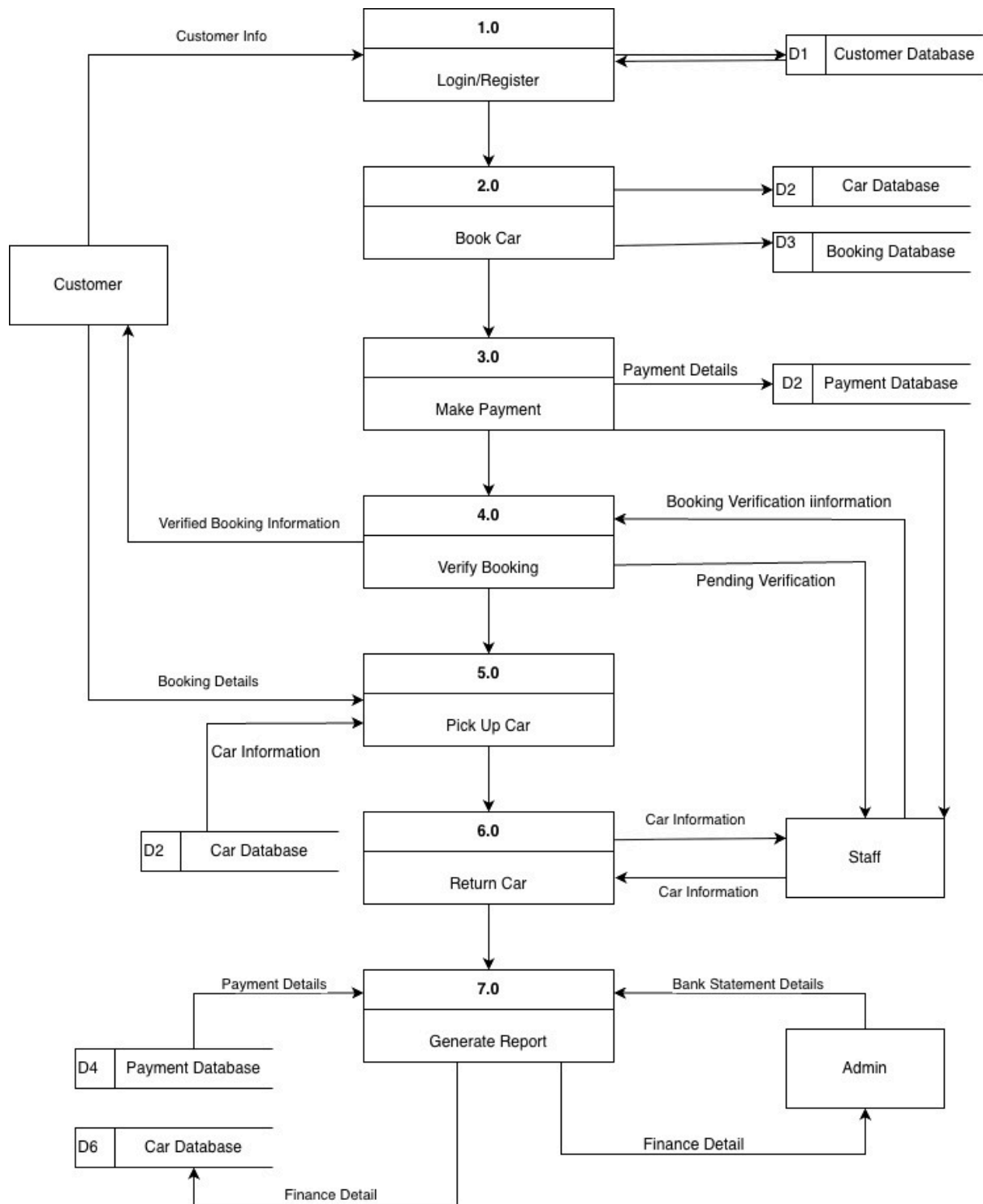


Figure 1: Data Flow Diagram (DFD) for the Proposed Car Rental System

3.0 DATA AND TRANSACTION REQUIREMENT

3.1 PROPOSED BUSINESS RULE

This section defines the business rules that determine how data is created, updated and managed in the proposed Hasta Travel & Tours Car Rental Management System. These rules ensure operations are consistent, prevent invalid transactions and automation on customer support.

3.1.1 Customer Registration and Verification

1. A customer must register an account before making a booking.
2. Each customer must have a unique Customer ID.
3. Customers must provide valid documents such as Identification Card or Passport, Matrics Card and Driving License.
4. Customer verification status will be recorded as either Pending, Approved or Rejected
5. Only customers with Approved verification status are allowed to make booking
6. Customers who are blacklisted are not allowed to create bookings until their blacklist status is removed by admin.

3.1.2 Vehicle Management

1. Each vehicle must have a unique Car ID and a unique Car Plate Number.
2. Vehicle status must be controlled by admin as Available, In Use and Maintenance
3. A vehicle can only be booked if its status is Available for the selected time.
4. A vehicle cannot be assignment to more than one booking in the same date and time
5. Vehicles that exceed the service mileage should be flagged by the system and set the status to Maintenance until resolved.

3.1.3 Booking

1. Each booking must have a unique Booking ID.
2. A booking must be created only by one verified customer.
3. A booking must be linked to only one vehicle
4. Booking must have a valid rental period.
5. The system should automatically change status to In Use once a customer has picked up the vehicle, and to Maintenance when the service mileage has exceeded.
6. A booking can only proceed to vehicle collection after payment is verified, signed agreement and necessary photo documents of the car are submitted.

3.1.4 Payment, Deposit and Refund

1. Payment types supported by the system are QR payment, Bank Transfer, and Cash
2. Payment receipt evidence must be recorded in the system
3. Deposit refunds should be processed after booking completion provided there are no outstanding penalties

3.1.5 Penalty Rules

1. Penalty records shall be listed to a booking and categorised as Late Return, Summon, Fuel Shortage, Damage or Cleanliness.
2. Customers with unpaid penalties may be blacklisted by the admin until the penalty is paid.

3.1.6 Car Collection, Return and Inspection

1. Customer should upload pictures of cars on all sides, signed agreement, fuel level before car collection
2. Customer should upload pictures of cars on all sides and fuel level after car collection
3. After car return and inspection done by the customer side, the system shall update vehicle status as Available unless maintenance is required.

3.1.7 Loyalty and Voucher

1. The system shall track loyalty stamps based on rental duration where 1 stamp

is awarded for every 9 rental hours.

2. Loyalty stamps and voucher eligibility shall be stored under the customer's account
3. Vouchers generated by admin must have a unique voucher code and expiry date.
4. Customers may redeem vouchers during booking payment and the system should record voucher usage to prevent reuse

3.1.8 Feedback and Issue Handling

1. Customers may submit feedback only for bookings that have status Completed
2. Vehicle issues reported in feedback shall be linked to the respective vehicle for tracking
3. Admin actions taken on issues must be recorded to support accountability and follow-up.

3.1.9 Reporting

1. Only admin are allowed to manage vehicles, verify payments, process refunds, and view reports and analytics
2. The system shall generate reports on bookings, revenue, vehicle utilisation and customer activity
3. Reports can be filtered by period to view the analytics.

3.2 PROPOSED DATA AND TRANSACTION

This section describes the key data requirements and transaction processes that will be supported by the proposed Hasta Travel & Tours Car Rental Management System. The proposed system will store, process, and manage structured data in a centralized database to ensure data integrity, consistency, and efficient transaction handling. The data and transactions are designed based on the business processes, functional requirements, and stakeholders' needs outlined in the previous sections.

3.2.1 Proposed Data Requirements

The proposed system will capture and store the following categories of data:

a) Customer Data

- Customer ID, name, email, phone number, residential college, and role (student/staff)
- Identification document details (IC/passport) and driving license information
- Verification status (pending, approved, rejected)
- Loyalty points and voucher records
- Deposit balance and blacklist status

b) Vehicle Data

- Car ID, model, brand, plate number, and category (e.g., sedan, MPV, SUV)
- Seating capacity, transmission type, and fuel type
- Current mileage and service limit status
- Vehicle status (available, booked, in use, maintenance, inactive)
- Pickup and drop-off locations

c) Booking Data

- Booking ID, booking date, rental start and end date/time
- Selected vehicle and assigned staff (if applicable)
- Booking status (pending, confirmed, completed, cancelled)
- Rental duration and total rental amount
- Deposit amount and payment status
- Before and after rental inspection photo records
- Digital agreement acceptance timestamp

d) Payment and Penalty Data

- Payment ID, booking ID, payment type (QR, bank transfer, cash)
- Payment amount, date, and receipt image
- Deposit record, refund status, and processing date
- Penalty records including late return fee, fuel penalty, damage/cleanliness charges
- Installment plan details (if applicable)

e) Inspection and Maintenance Data

- Inspection ID, booking ID, inspector (staff/admin)
- Pre-rental and post-rental condition reports
- Recorded damages, fuel level, and cleanliness status
- Maintenance alerts when mileage exceeds service limit

f) Feedback and Issue Data

- Feedback ID, customer ID, booking ID
- Customer ratings and comments
- Reported issues linked to specific vehicles
- Admin/staff actions taken in response to issues

3.2.2 Proposed Transaction Requirements

The system will support the following core transactions:

a) User Registration and Verification Transaction

- Customers register using institutional credentials
- Upload identification and driving license documents
- Admin verifies and updates user status
- System records agreement acceptance and activates account

b) Login and Blacklist Transaction

- System authenticates verified users
- System blocks access for blacklisted users until penalties are settled
- Admin can update blacklist reasons and status

c) Vehicle Management Transaction (Admin/Staff)

- Add, update, deactivate vehicle records
- Update vehicle status in real time
- System automatically flags vehicles exceeding service mileage

d) Booking Transaction (Customer)

- Customer browses available vehicles in real time
- Customer selects rental period and confirms booking
- System reserves vehicle and generates booking record
- Customer uploads key handover and inspection photos
- Customer digitally signs rental agreement
- System updates booking status accordingly

e) Payment and Penalty Transaction

- Customer uploads payment receipt
- Admin verifies payment and updates status
- System records deposit, rental payment, and penalties
- System calculates penalties automatically:
 - RM30 per hour for late return

- RM10 per fuel bar shortage
- Additional charges for damage/cleanliness
- System processes deposit refund or retention within 10 working days

f) Car Collection and Return Transaction

- Staff conducts pre-rental inspection and records data
- System logs car handover details
- Upon return, staff records post-rental inspection
- System updates vehicle status and calculates penalties if applicable

g) Loyalty and Voucher Transaction

- System tracks rental hours (1 stamp per 9 hours)
- Admin generates vouchers when criteria are met
- Customer redeems vouchers during checkout
- System updates loyalty records accordingly

h) Reporting and Analytics Transaction (Admin)

- System generates real-time dashboard statistics
- Admin can view booking trends, revenue, and vehicle utilization
- Reports can be filtered by residential college, period, or vehicle type

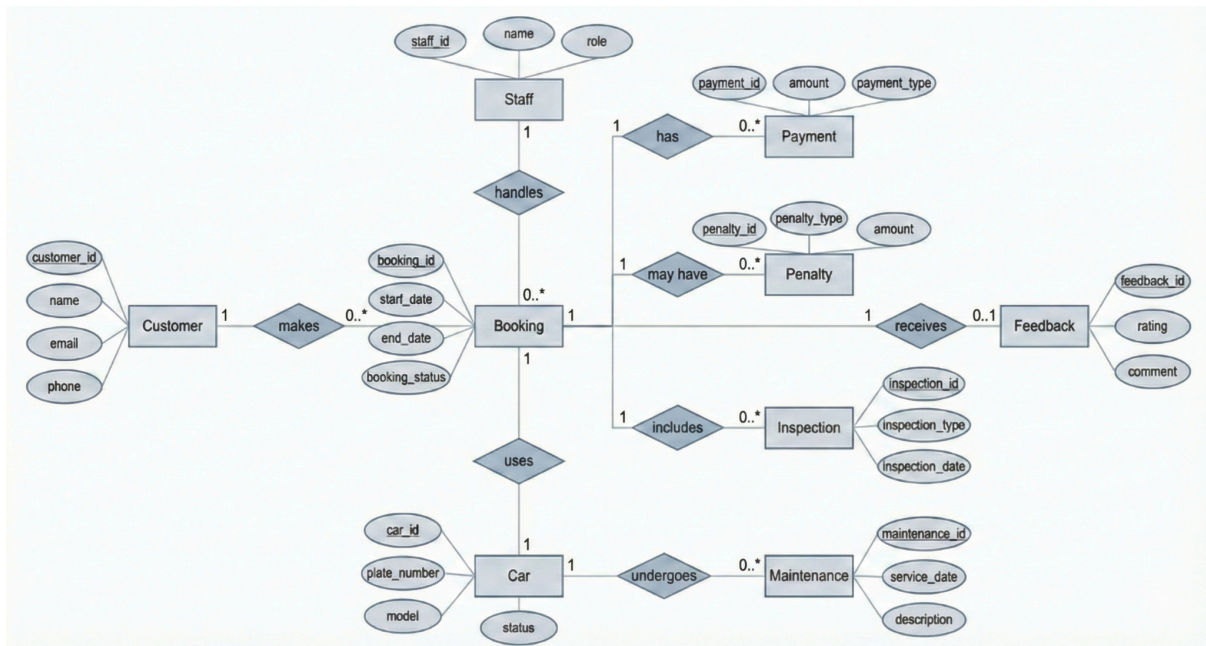
3.2.3 Summary

The proposed data and transaction structure ensures that all critical business processes of Hasta Travel & Tours are supported in a centralized, reliable, and scalable system. By integrating customer, vehicle, booking, payment, inspection, and feedback data into a single platform, the system enhances data consistency, operational efficiency, and decision-making capabilities while improving the overall user experience for both customers and staff.

4.0 DATABASE CONCEPTUAL DESIGN

4.1 CONCEPTUAL ERD

This conceptual Entity-Relationship Diagram (ERD) visualizes the high-level data structure of a Car Rental Management System, centering on the Booking entity as the operational hub. It illustrates the core business flow where a Customer makes a reservation that is handled by Staff and utilizes a specific Car. The diagram further details the rental lifecycle by connecting bookings to financial elements like Payment and potential Penalty entities, as well as operational procedures involving Inspection and customer Feedback. Additionally, it accounts for asset management by linking the Car entity directly to its Maintenance history to track vehicle upkeep.

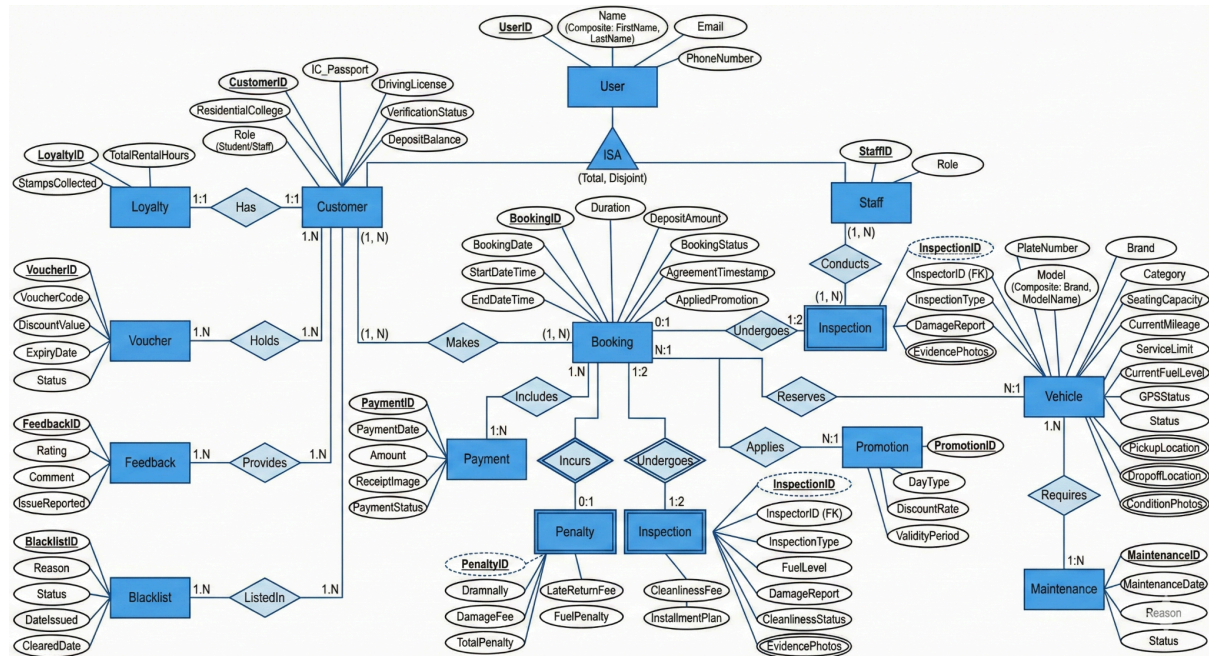


4.2 ENHANCED ERD (EERD)

This Enhanced Entity-Relationship Diagram (EERD) builds upon the conceptual ERD by incorporating advanced modeling features such as specialization/generalization, weak entities, composite attributes, and multivalued attributes. The Booking entity remains the central hub of the system, with dependent relationships to Payment, Penalty, and Inspection modeled as weak entities.

The EERD introduces a generalized User entity that is specialized into Customer and Staff through a total and disjoint ISA relationship. It further refines data representation by introducing Blacklist for penalty enforcement and Promotion for dynamic pricing. The model

also links Vehicle to Maintenance to track service history, and integrates Loyalty and Voucher to represent the rewards system. Overall, the EERD provides a more detailed, structured, and semantically rich representation of the system's data requirements.



5.0 DATA DICTIONARY

This data dictionary describes the data elements defined in both the Conceptual ERD and the Enhanced ERD (EERD) for the Hasta Travel & Tours Car Rental Management System. It documents all entities, attributes, and specialization structures shown in the ERD diagrams, without including physical database or implementation-specific details.

Supertype: User

Description: Represents all users who interact with the system. This entity is specialized into

Customer and Staff using a total and disjoint ISA relationship.

Attribute	Description
UserID	Unique identifier for each user
Name	Composite attribute consisting of FirstName and LastName
Email	User email address
PhoneNumber	User contact number

Subtype: Customer

Description: Represents customers who register, make bookings, and rent vehicles.

Attribute	Description
CustomerID	Unique identifier for customer
IC_Passport	Customer identification document number
DrivingLicense	Customer driving license information
VerificationStatus	Status of customer verification
DepositBalance	Current deposit balance held by the system
ResidentialCollege	Residential college of customer
Role	Customer role (e.g., student or staff)

Subtype: Staff

Description: Represents staff members responsible for handling bookings, inspections, and verification.

Attribute	Description
StaffID	Unique identifier for staff
Role	Role of staff in the system

Entity: Loyalty

Description: Stores customer loyalty information accumulated through rental activities.

Attribute	Description
LoyaltyID	Unique identifier for loyalty record
TotalRentalHours	Total rental hours accumulated by customer
StampsCollected	Number of loyalty stamps collected

Entity: Voucher

Description: Represents discount vouchers that can be redeemed by customers.

Attribute	Description
VoucherID	Unique identifier for voucher
VoucherCode	Voucher redemption code
DiscountValue	Discount amount or rate

ExpiryDate	Voucher expiration date
Status	Current voucher status

Entity: Blacklist

Description: Stores blacklist records for customers who violate system rules.

Attribute	Description
BlacklistID	Unique identifier for blacklist record
Reason	Reason for blacklisting
Status	Current blacklist status
DateIssued	Date blacklist was issued
ClearedDate	Date blacklist was cleared

Entity: Booking

Description: Represents a vehicle booking transaction and serves as the central entity of the system.

Attribute	Description
BookingID	Unique identifier for booking
BookingDate	Date booking was created
StartDateTime	Rental start date and time
EndDateTime	Rental end date and time

Duration	Total rental duration
DepositAmount	Deposit amount required
BookingStatus	Current booking status
AgreementTimestamp	Timestamp of agreement acceptance
AppliedPromotion	Applied promotion identifier

Entity: Payment

Description: Represents payment records associated with a booking.

Attribute	Description
PaymentID	Unique identifier for payment
PaymentDate	Date payment was made
Amount	Payment amount
ReceiptImage	Proof of payment
PaymentStatus	Status of payment

Entity: Penalty

Description: Represents penalties incurred due to booking violations.

Attribute	Description
PenaltyID	Unique identifier for penalty
LateReturnFee	Fee charged for late return

FuelPenalty	Penalty for insufficient fuel
DamageFee	Fee for vehicle damage
TotalPenalty	Total penalty amount

Entity: Inspection

Description: Stores inspection records conducted before and after vehicle rental.

Attribute	Description
InspectionID	Unique identifier for inspection
InspectorID	Identifier of staff conducting inspection
InspectionType	Type of inspection
FuelLevel	Fuel level recorded during inspection
DamageReport	Description of damages found
CleanlinessStatus	Cleanliness condition of vehicle
EvidencePhotos	Photos taken as inspection evidence

Entity: Promotion

Description: Represents promotional offers applied to bookings.

Attribute	Description
PromotionID	Unique identifier for promotion

DayType	Applicable day type
DiscountRate	Discount rate applied
ValidityPeriod	Promotion validity period

Entity: Vehicle

Description: Represents vehicles available for rental.

Attribute	Description
PlateNumber	Vehicle registration number
Brand	Vehicle brand
Model	Composite attribute consisting of Brand and ModelName
Category	Vehicle category
SeatingCapacity	Number of seats available
CurrentMileage	Current mileage of vehicle
ServiceLimit	Mileage limit before service is required
CurrentFuelLevel	Current fuel level
GPSStatus	GPS availability status
Status	Vehicle availability status
PickupLocation	Vehicle pickup location
DropoffLocation	Vehicle drop-off location
ConditionPhotos	Photos showing vehicle condition

Entity: Maintenance

Description: Stores maintenance and service records for vehicles.

Attribute	Description
MaintenanceID	Unique identifier for maintenance
MaintenanceDate	Date maintenance was performed
Reason	Reason for maintenance
Status	Maintenance status

6.0 Summary

This report completely covered on the conceptual database design for Hasta Travel & Tours Car Rental Management System. The To-Be Data Flow Diagram (DFD) is developed to show how data moves through the important processes mentioned such as registration, booking, payment, verification and many more. Next, the business rules were defined to control those processes in order for all the actions to follow the organisational constraints and policies. The data & transactional requirements are defined to ensure all necessary information is available to suppose the functions in the system. The conceptual ERD and enhanced EERD is created to represent high level structure of the database with relationships among entities. This report provides a strong foundation for logical database development which helps better decision making for the management of Hasta Travel & Tours to improve their customer experience.